

Topological Quantum Computation

George Toh

11/6/2017

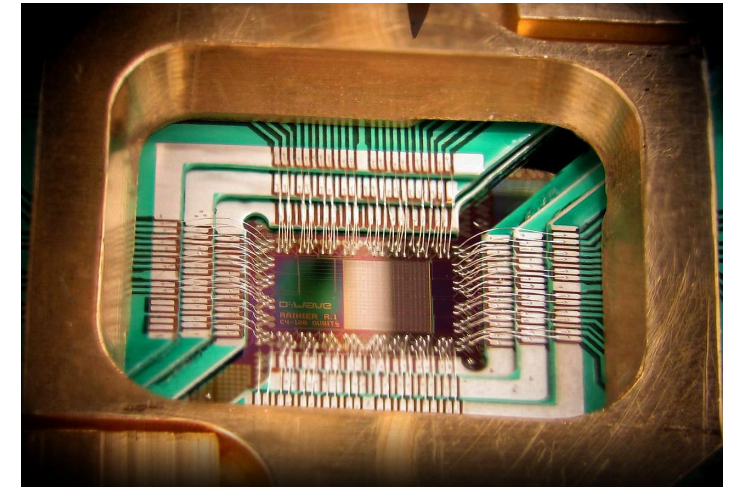
Contents

- Quantum Computing
- Comparison of QC vs TQC
- Topological Quantum Computation
- How to implement TQC?
- Examples, progress
- Industry investment
- Future outlook

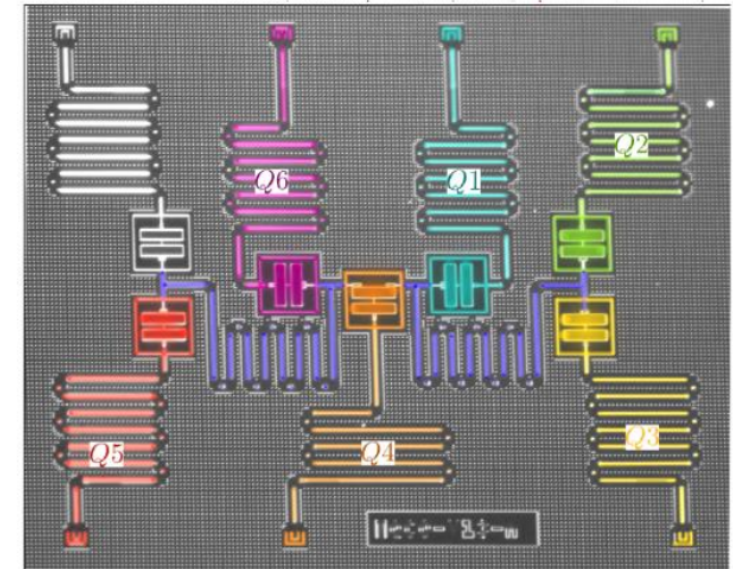
Timeline of Quantum Computing

- 1980s – Benioff, Manin, Feynman and Deutsch
- 1994 – Shor's Algorithm (factoring)
- 1995 – First quantum logic gate (trapped ions)
- **1997 – Kitaev describes Topological QC**
- 2001 – 7 bit NMR QC + Shor's Algo: 15
- 2011 – D-Wave 128-bit Quantum Annealer
- **2013 – Evidence of non-Abelian anyons**
- 2016 – Google & UCSB simulate hydrogen molecule
- 2016 – IBM QX 5 bit QC (online!)
- 2017 – IBM QX 16 bit QC

D-Wave systems chip



IBM 7-bit chip



Quantum Computing

- Applications
 - Quantum factoring (Shor's Algorithm)
 - Efficient search (Grover's Algorithm)
 - Quantum simulation
 - Quantum supremacy (>56 Qubits)
- Qubits
 - Trapped ions, superconducting loops, quantum dots



Pioneers in TQC

- Alexei Kitaev (Caltech)
 - 1997 - First proposed Topological quantum computation
 - 2002 - Proved that a TQC can do any quantum computation (with Freedman and others)
- Michael Freedman (UC Santa Barbara)
 - 2005 - Proposed that a quantum Hall device could realize a topological qubit
 - Director of Microsoft Station Q at UCSB



A. Yu. Kitaev

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117940, Kosygina St. 2*

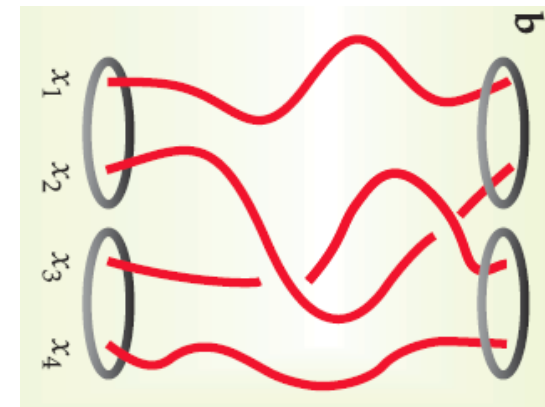
Comparison – Why TQC?

Quantum Computing

- Qubits
- Quantum decoherence (Noise)
- Extremely fragile
- Need to be shielded from environmental perturbations

Topological Quantum Computing

- Topological qubits based on non-Abelian anyons
- These two-dimensional quasiparticles form braids which are much more stable



Non-Abelian anyons

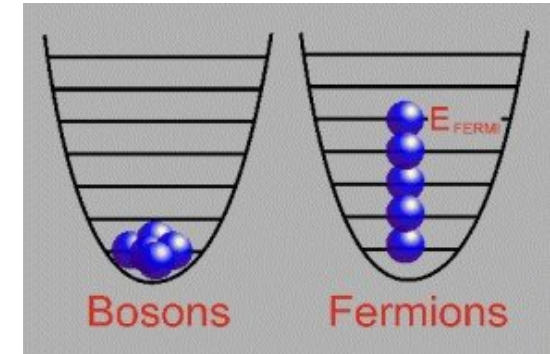
Fermion and Boson

- Abelian
- Fermi-Dirac statistics or
- Bose-Einstein statistics

Standard Model of Elementary Particles

three generations of matter (fermions)

	I	II	III	
mass	$\sim 2.4 \text{ MeV}/c^2$	$\sim 1.275 \text{ GeV}/c^2$	$\sim 172.44 \text{ GeV}/c^2$	0
charge	$2/3$	$2/3$	$2/3$	0
spin	$1/2$	$1/2$	$1/2$	1
	u up	c charm	t top	g gluon
QUARKS	d down	s strange	b bottom	γ photon
	$\sim 4.8 \text{ MeV}/c^2$	$\sim 95 \text{ MeV}/c^2$	$\sim 4.18 \text{ GeV}/c^2$	0
	$-1/3$	$-1/3$	$-1/3$	0
	$1/2$	$1/2$	$1/2$	1
	e electron	μ muon	τ tau	Z Z boson
LEPTONS	$\sim 0.511 \text{ MeV}/c^2$	$\sim 105.67 \text{ MeV}/c^2$	$\sim 1.7768 \text{ GeV}/c^2$	$\sim 91.19 \text{ GeV}/c^2$
	-1	-1	-1	0
	$1/2$	$1/2$	$1/2$	1
	ν_e electron neutrino	ν_μ muon neutrino	ν_τ tau neutrino	W W boson
	$\sim 2.2 \text{ eV}/c^2$	$\sim 1.7 \text{ MeV}/c^2$	$\sim 35.5 \text{ MeV}/c^2$	$\sim 80.39 \text{ GeV}/c^2$
	0	$1/2$	$1/2$	1
	$1/2$	$1/2$	$1/2$	1
				H Higgs
				$\sim 125.09 \text{ GeV}/c^2$
				0
				0
				1
				SCALAR BOSONS
				GAUGE BOSONS

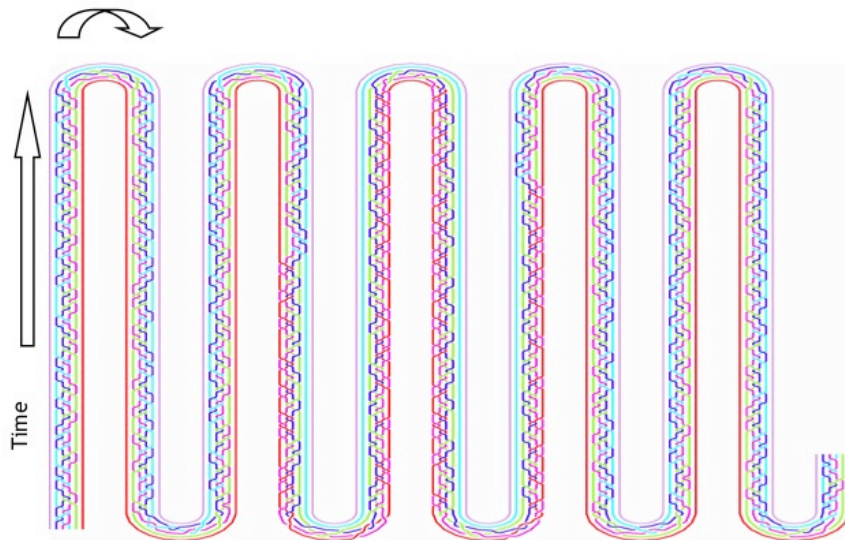


Anyon

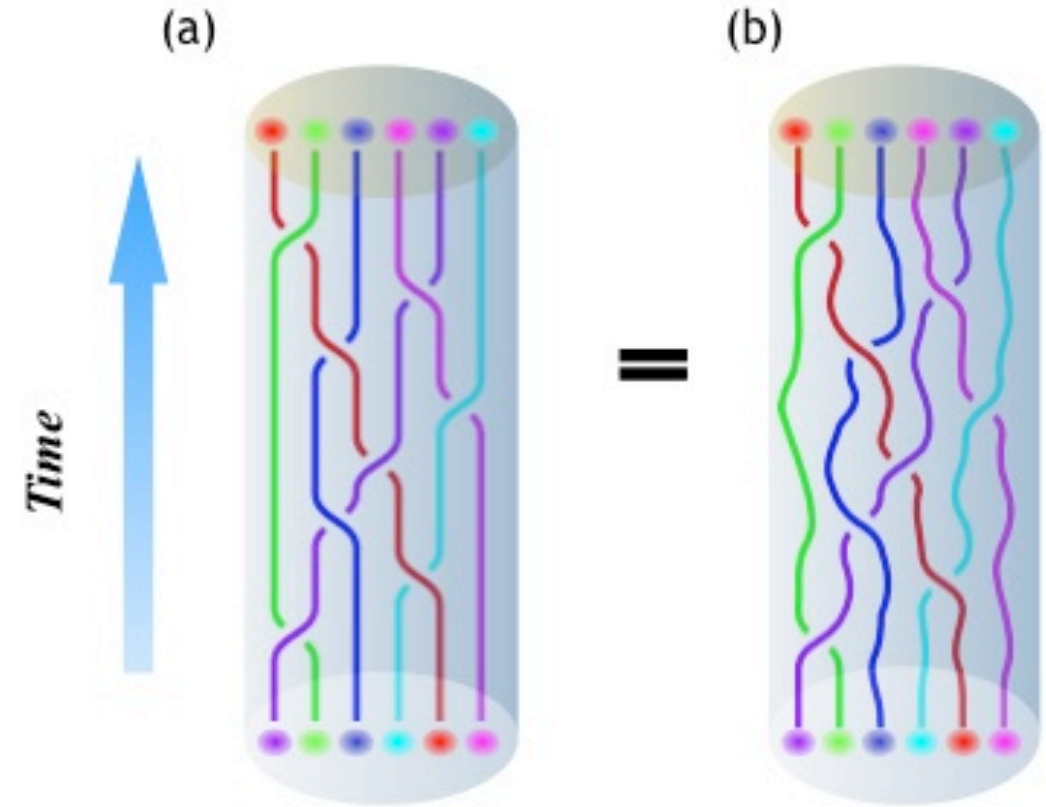
- 2-Dimensional quasiparticle
- Non-Abelian
- Possible to be somewhere in between F-D and B-E statistics
- Particle interchange: phase change does not have to be +1 or -1

Topological Quantum Computation

- Non-Abelian anyons
- Braid = quantum operation
- Fault-tolerant

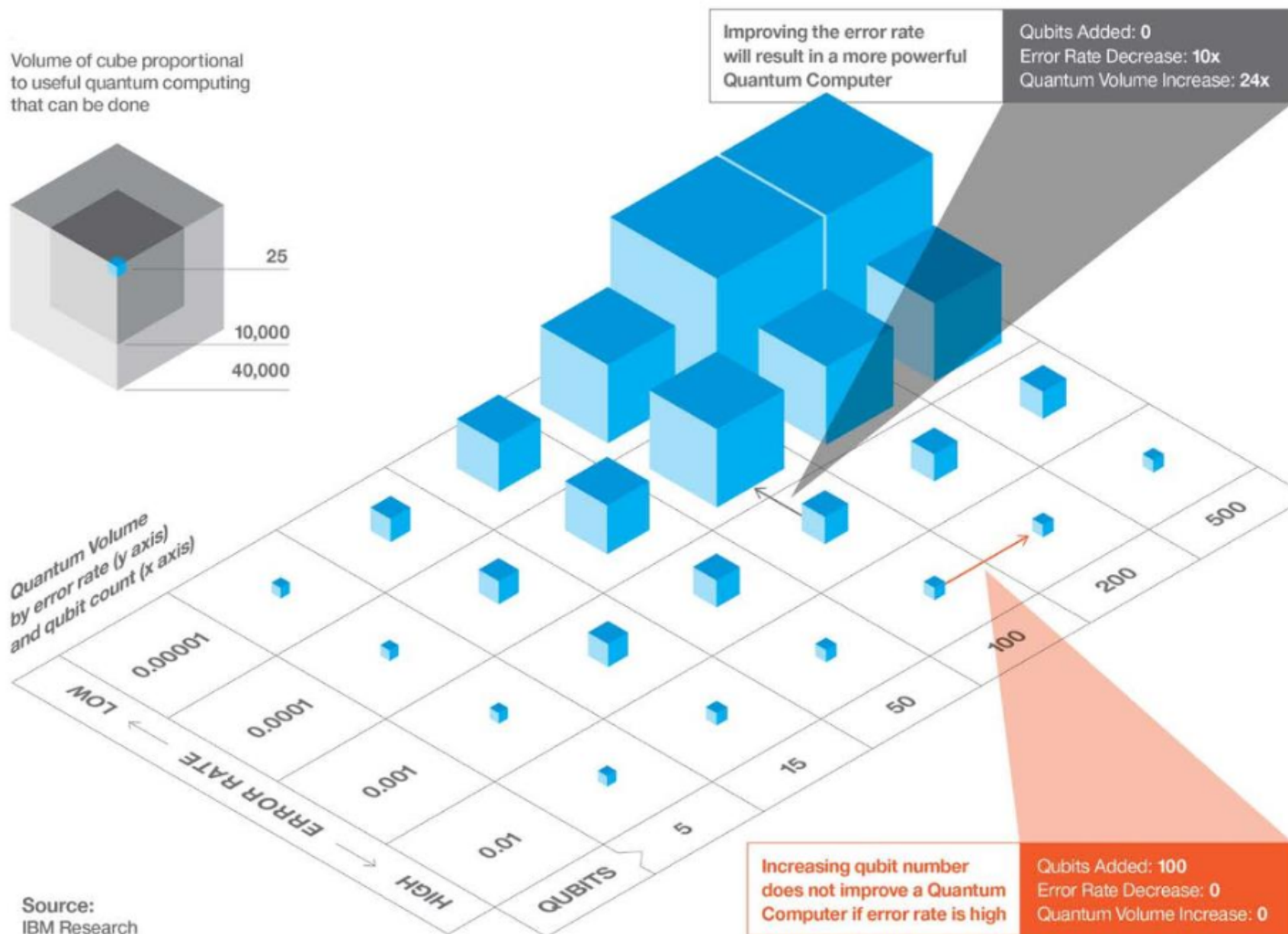


Controlled KNOT



(a) and (b) are topologically identical

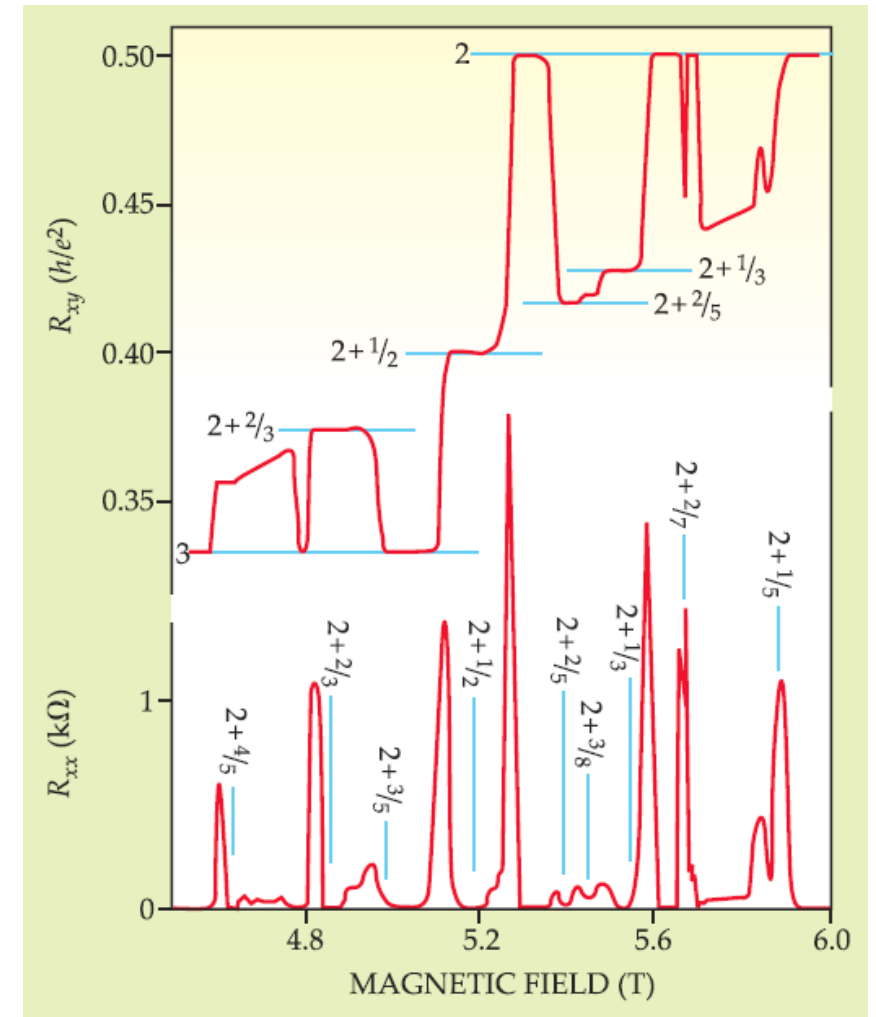
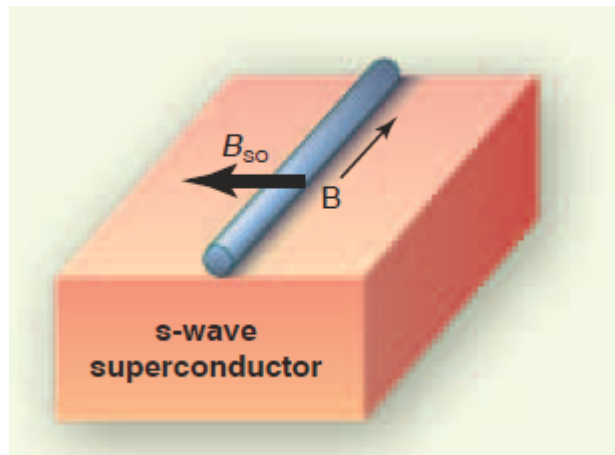
Quantum Volume



Source:
IBM Research

Candidates for non-Abelian anyons

- Nanowire on superconductor
 - Majorana zero modes
- Fractional Quantum Hall State at (filling factor) $\nu = 5/2$

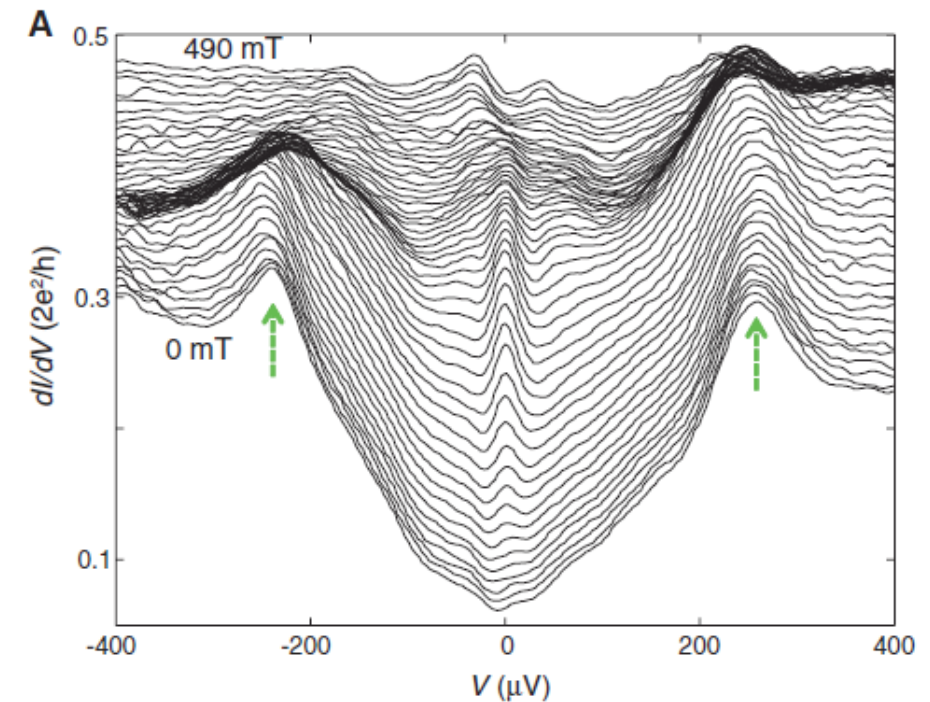
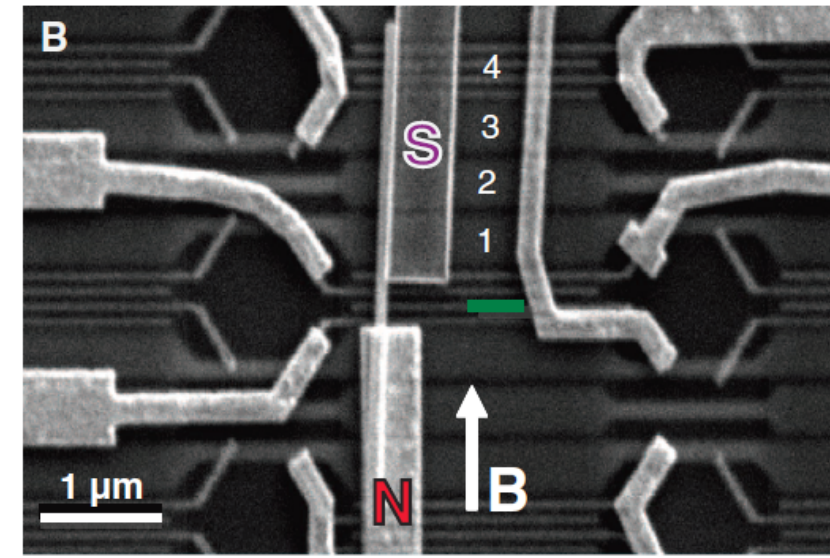
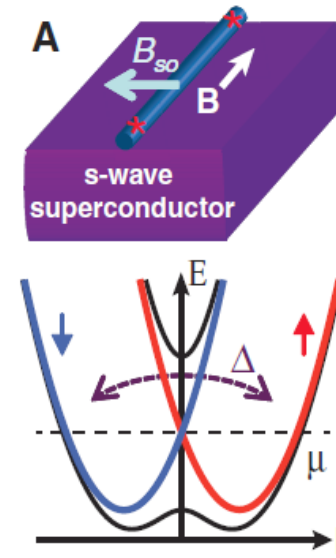


Hybrid Superconductor-Semiconductor Nanowire Devices

- Semiconductor InSb nanowire in contact with an ordinary superconductor Nb
- Zero energy peaks show up only when a finite external magnetic field is applied to the wire

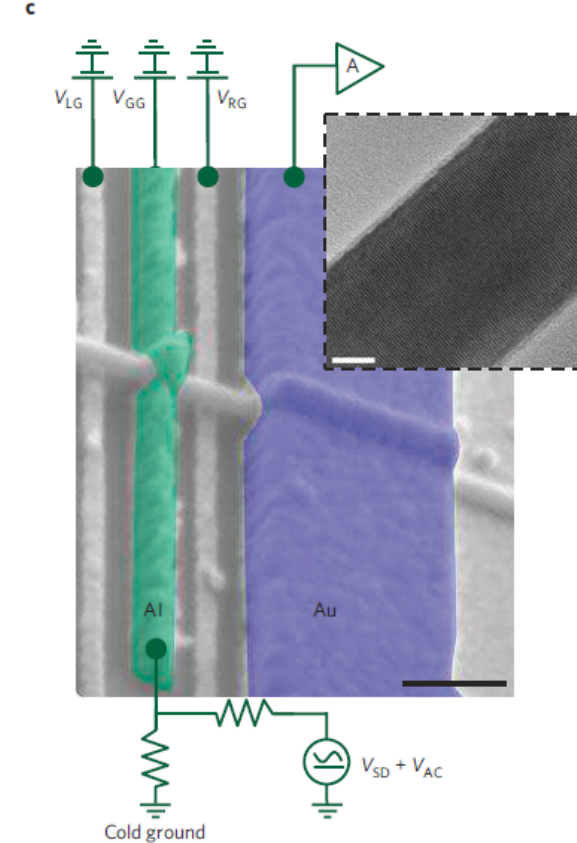
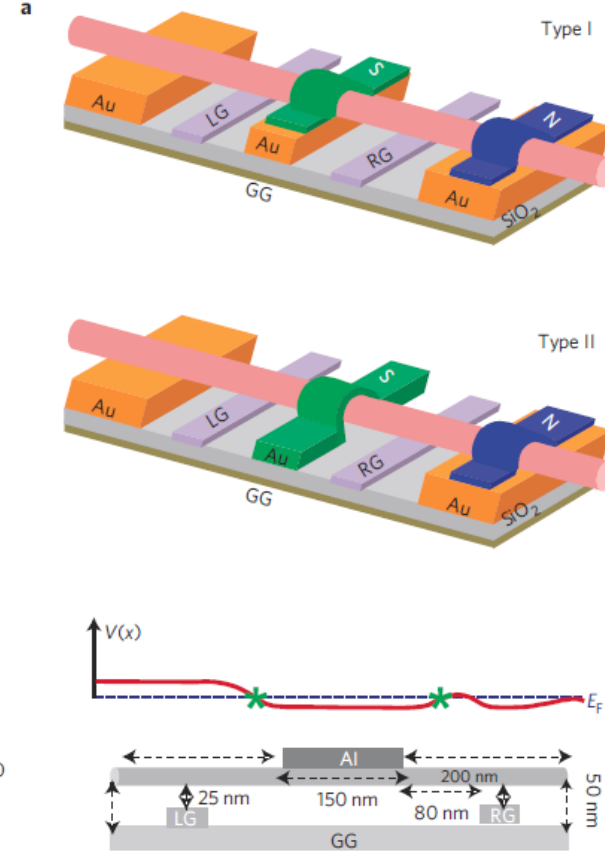
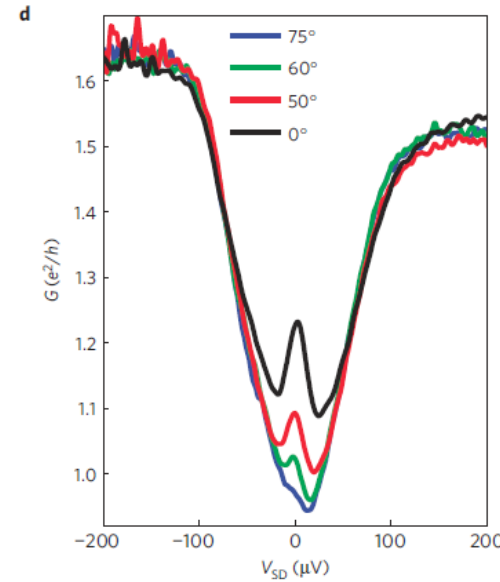
Signatures of Majorana Fermions in Hybrid Superconductor-Semiconductor Nanowire Devices

V. Mourik,^{1*} K. Zuo,^{1*} S. M. Frolov,¹ S. R. Plissard,² E. P. A. M. Bakkers,^{1,2} L. P. Kouwenhoven^{1†}



Hybrid Superconductor-Semiconductor Nanowire Devices

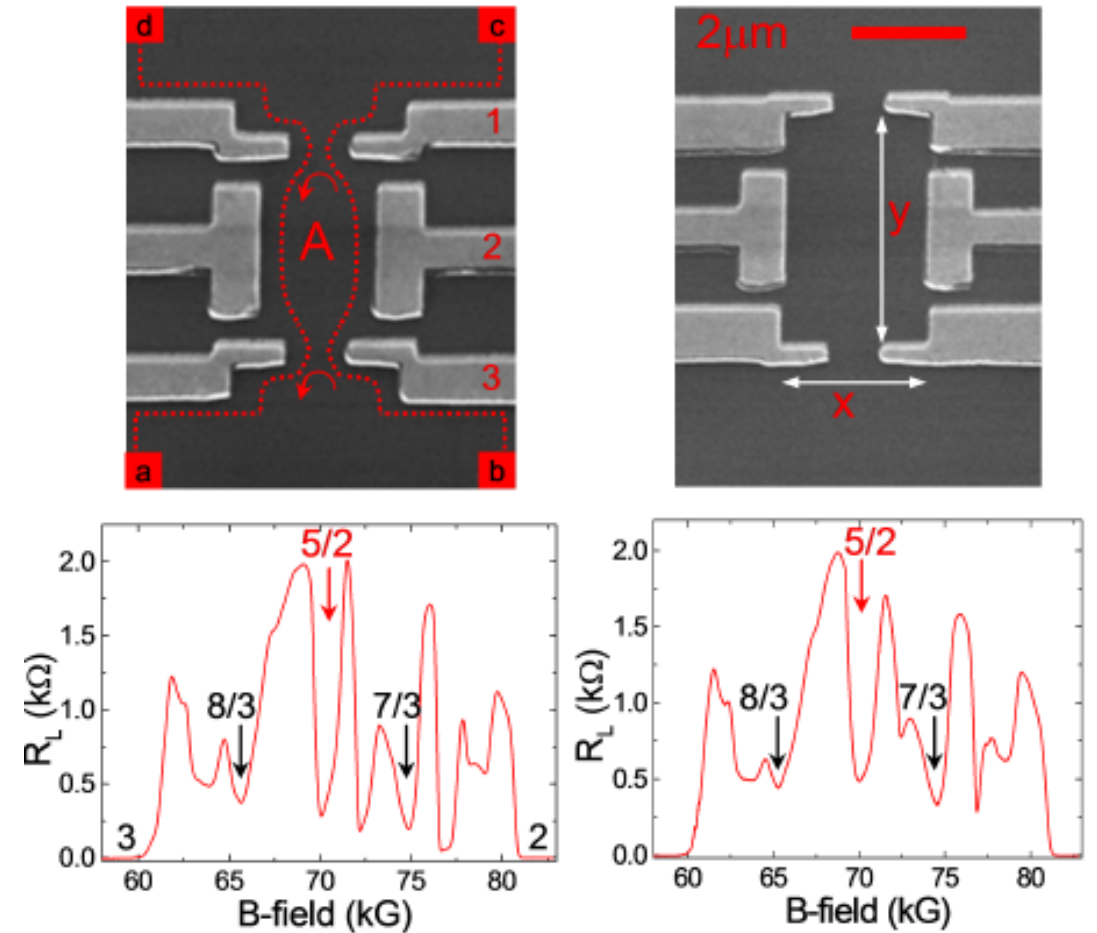
- Similar experiment to Delft, published a few months later in 2012
- InAs nanowire on Al superconductor



Zero-bias peaks and splitting in an Al-InAs nanowire topological superconductor as a signature of Majorana fermions

Experimental evidence of non-Abelian anyons

- Fabry-Perot Interferometer made of GaAs/AlGaAs quantum well with a SiN layer
- Period of A-B oscillation indicative of non-Abelian nature of $\nu=5/2$ state



Magnetic field-tuned Aharonov-Bohm oscillations and evidence for non-Abelian anyons at $\nu = 5/2$

R. L. Willett,¹ C. Nayak,^{2,3} K. Shtengel,^{2,4} L. N. Pfeiffer,⁵ and K. W. West⁵

¹*Bell Laboratories, Alcatel-Lucent, Murray Hill, New Jersey 07974, USA*

²*Microsoft Research, Station Q, Elings Hall, University of California, Santa Barbara, California 93106, USA*

³*Department of Physics, University of California, Santa Barbara, California 93106, USA*

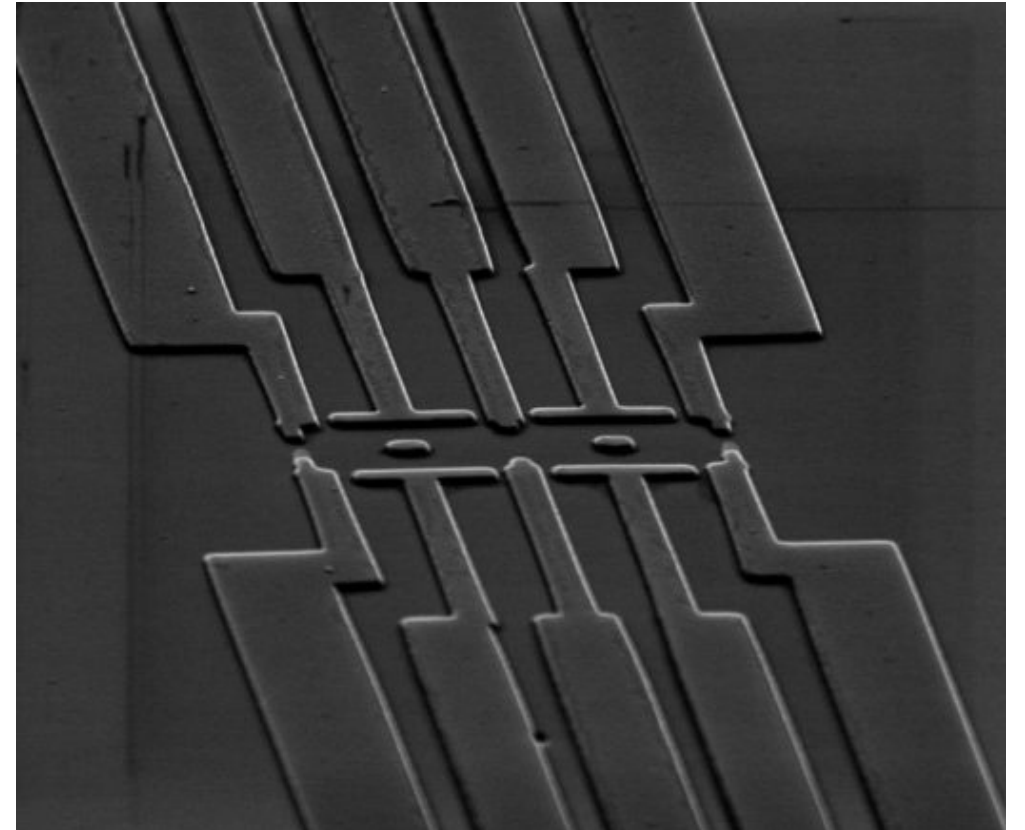
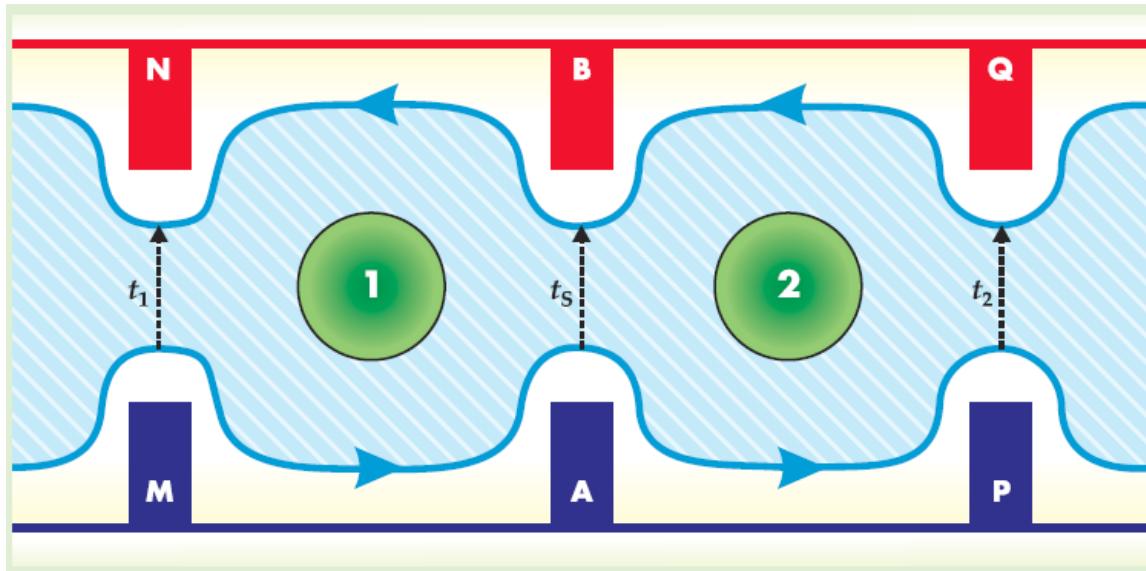
⁴*Department of Physics and Astronomy, University of California, Riverside, California 92521, USA*

⁵*Department of Electrical Engineering, Princeton University, Princeton, New Jersey 08544, USA*

(Dated: January 15, 2013)

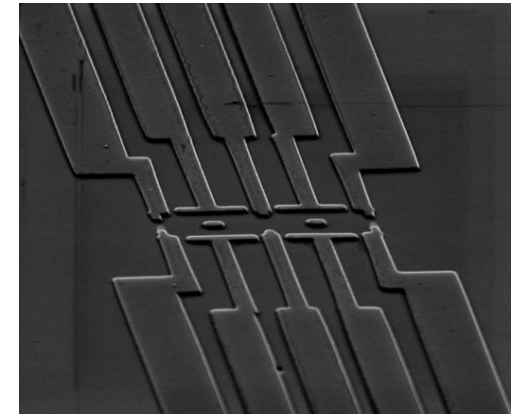
Experimental evidence of non-Abelian anyons

- Next step is to implement and test a two interferometer setup



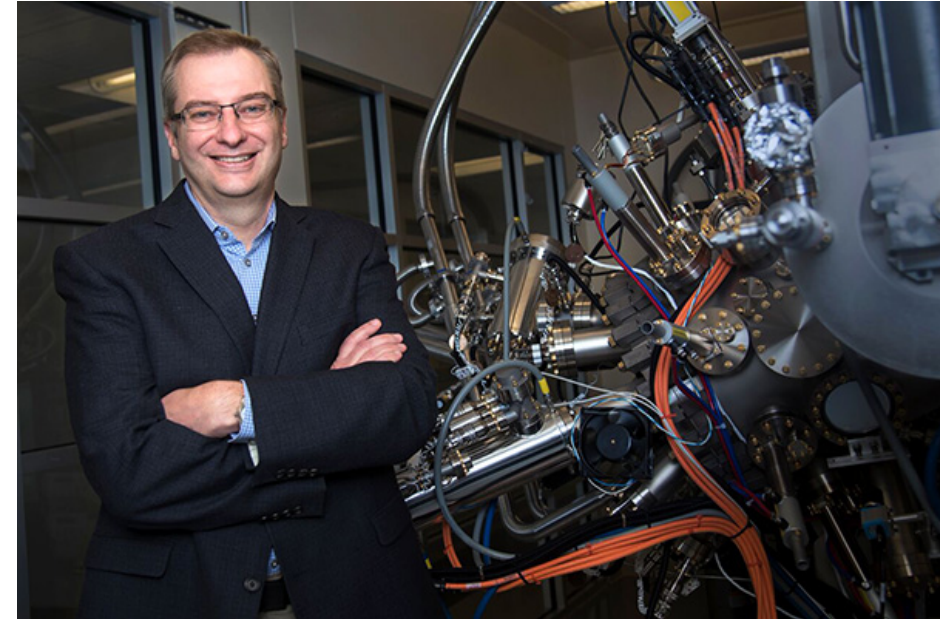
Investments in TQC

- Microsoft: Station Q
 - UCSB, Prof Michael Freedman, Theory
 - TU Delft and University of Sydney, Experimental
 - Purdue, Prof Manfra, Experimental
- Intel and QuTech
 - Delft University
 - US\$50 million
- Bell Labs
 - Robert Willett



Purdue – Microsoft Station Q

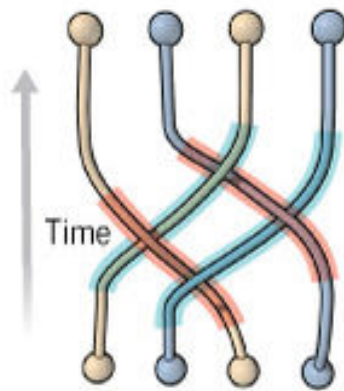
- Purdue University and Microsoft Corp. have signed a multi-year agreement to develop a useable quantum computer (May 2017)
- Grow and study ultra-pure semiconductors and hybrid systems of semiconductors and superconductors
- Manfra's group has expertise in a technique called molecular beam epitaxy, used to build low-dimensional electron systems that form the basis for quantum bits, or qubits.



Conclusion

- Quantum computers are hot due to the potential applications
- TQC offer a way to solve the decoherence/noise problem
- Microsoft is betting big on TQC
- TQC not yet experimentally demonstrated

Topological qubits



Quasiparticles can be seen in the behavior of electrons channeled through semiconductor structures. Their braided paths can encode quantum information.

Longevity (seconds)	N/A
Logic success rate	N/A
Number entangled	N/A

Company support

Microsoft, Bell Labs

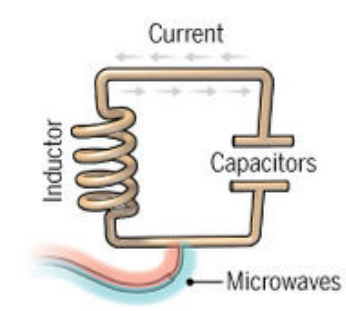
+ Pros

Greatly reduce errors.

- Cons

Existence not yet confirmed.

Superconducting loops



A resistance-free current oscillates back and forth around a circuit loop. An injected microwave signal excites the current into super-position states.

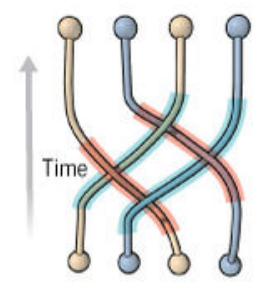
Longevity (seconds)	0.000005
Logic success rate	99.4%
Number entangled	9

Company support

Google, IBM, Quantum Circuits

- Pros**
Fast working. Build on existing semiconductor industry.
- Cons**
Collapse easily and must be kept cold.

Topological qubits



Quasiparticles can be seen in the behavior of electrons channeled through semiconductor structures. Their braided paths can encode quantum information.

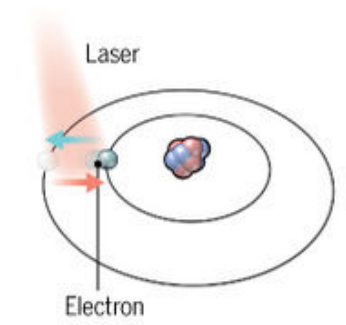
Longevity (seconds)	N/A
Logic success rate	N/A
Number entangled	N/A

Company support

Microsoft, Bell Labs

- Pros**
Greatly reduce errors.
- Cons**
Existence not yet confirmed.

Trapped ions



Electrically charged atoms, or ions, have quantum energies that depend on the location of electrons. Tuned lasers cool and trap the ions, and put them in super-position states.

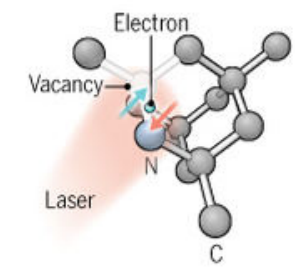
Longevity (seconds)	>1000
Logic success rate	99.9%
Number entangled	14

Company support

ionQ

- Pros**
Very stable. Highest achieved gate fidelities.
- Cons**
Slow operation. Many lasers are needed.

Diamond vacancies



A nitrogen atom and a vacancy add an electron to a diamond lattice. Its quantum spin state, along with those of nearby carbon nuclei, can be controlled with light.

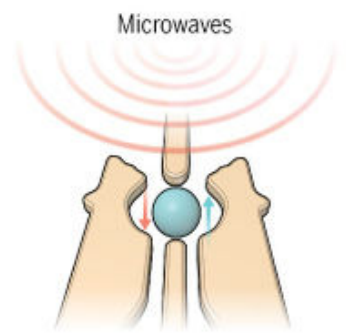
Longevity (seconds)	10
Logic success rate	99.2%
Number entangled	6

Company support

Quantum Diamond Technologies

- Pros**
Can operate at room temperature.
- Cons**
Difficult to entangle.

Silicon quantum dots



These "artificial atoms" are made by adding an electron to a small piece of pure silicon. Microwaves control the electron's quantum state.

Longevity (seconds)	0.03
Logic success rate	~99%
Number entangled	2

Company support

Intel

- Pros**
Stable. Build on existing semiconductor industry.
- Cons**
Only a few entangled. Must be kept cold.

Note: Longevity is the record coherence time for a single qubit superposition state, logic success rate is the highest reported gate fidelity for logic operations on two qubits, and number entangled is the maximum number of qubits entangled and capable of performing two-qubit operations.

Dirac notation

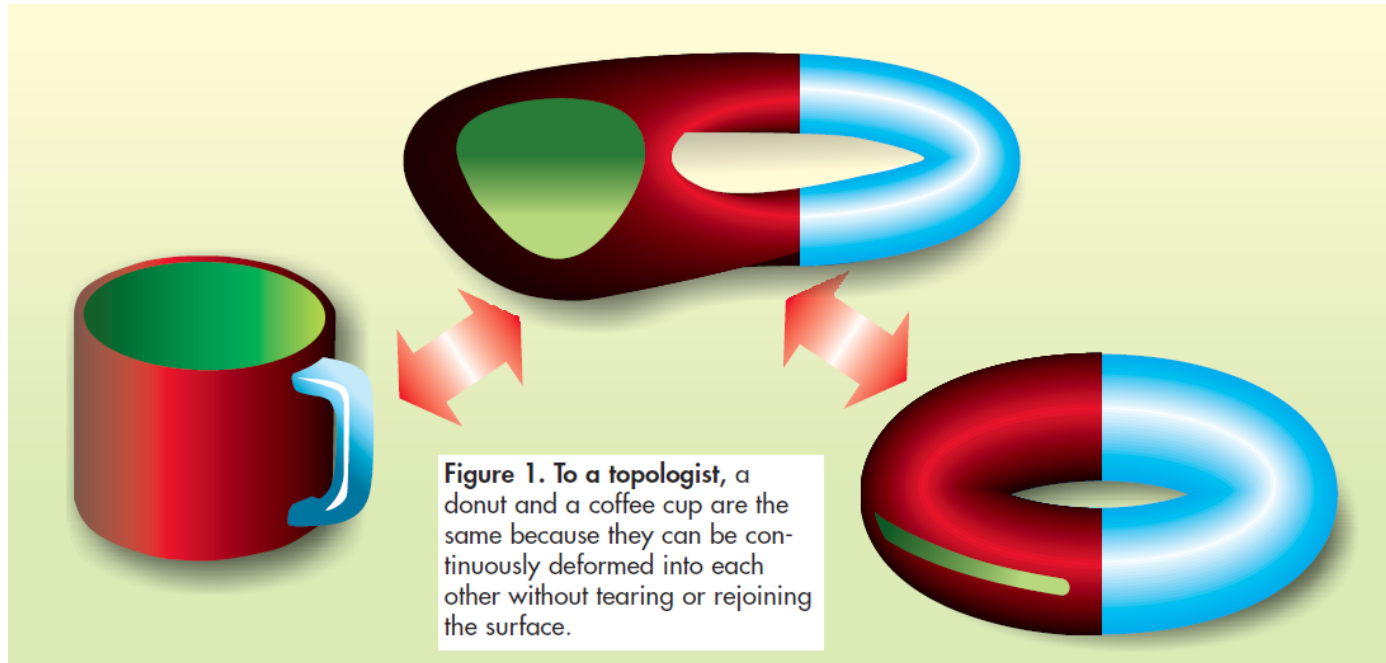
Bosons: $|\psi_1\psi_2\rangle = +|\psi_1\psi_2\rangle$

Fermions: $|\psi_1\psi_2\rangle = -|\psi_1\psi_2\rangle$

Anyons: $|\psi_1\psi_2\rangle = e^{i\theta}|\psi_1\psi_2\rangle$

What is Topology?

- 2016 Nobel Prize in Physics
- Topology: In math, describes a property that remains unchanged under smooth continuous deformation



The Nobel Prize in Physics 2016



Photo: A. Mahmoud
David J. Thouless
Prize share: 1/2



Photo: A. Mahmoud
F. Duncan M. Haldane
Prize share: 1/4



Photo: A. Mahmoud
J. Michael Kosterlitz
Prize share: 1/4

The Nobel Prize in Physics 2016 was awarded with one half to David J. Thouless, and the other half to F. Duncan M. Haldane and J. Michael Kosterlitz *"for theoretical discoveries of topological phase transitions and topological phases of matter"*.

Topological invariance:

To change the topological state, a large perturbation making another hole is required